Lab 1 - Diodes

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ECE375

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Introduction

This lab goes over the characteristics of the diode. It starts with generating current-voltage curves (IV curves), and from there the parameters of the diode (like saturation current, series resistance, and diode ideal factor).

All code and full tables can be found on my Github page here. This is for transparency on what code I used, what data I'm plotting, and version control for how it developed over time.

Alternatively accessed with this link: https://github.com/arshbhamla/ece375-semiconductor-lab

1. Components

- 1N4148 Diode
    - 200mA max current rating
- 2 Banana to Alligator cables
- 1 Banana to Banana cable
- 1 BNC to Alligator cable
- EDU36311A DC Power Supply
- EDU34450A Multimeter
- DSIX1292G Oscilloscope

2. Circuit Diagram

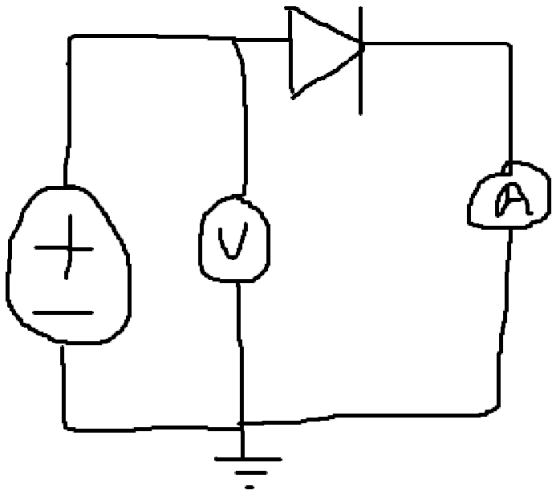


Figure 1. Circuit Diagram

For the voltmeter, we are using the voltage reading of the DC power supply, and the ammeter uses the multimeter.

3. Measurement Values

Set of sample values to track the trend. Link to the full dataset is below this table

Voltage	Current (mA)
0	0.0001
0.1	0.0005
0.2	0.0007
0.3	0.0022
0.4	0.0111
0.5	0.08701
0.6	0.74
0.7	5.102
0.8	19.7
0.9	45.351
1	78.747

Table 1. Table of Measured Values

Click here for the full table

4. Experimental IV Graphs

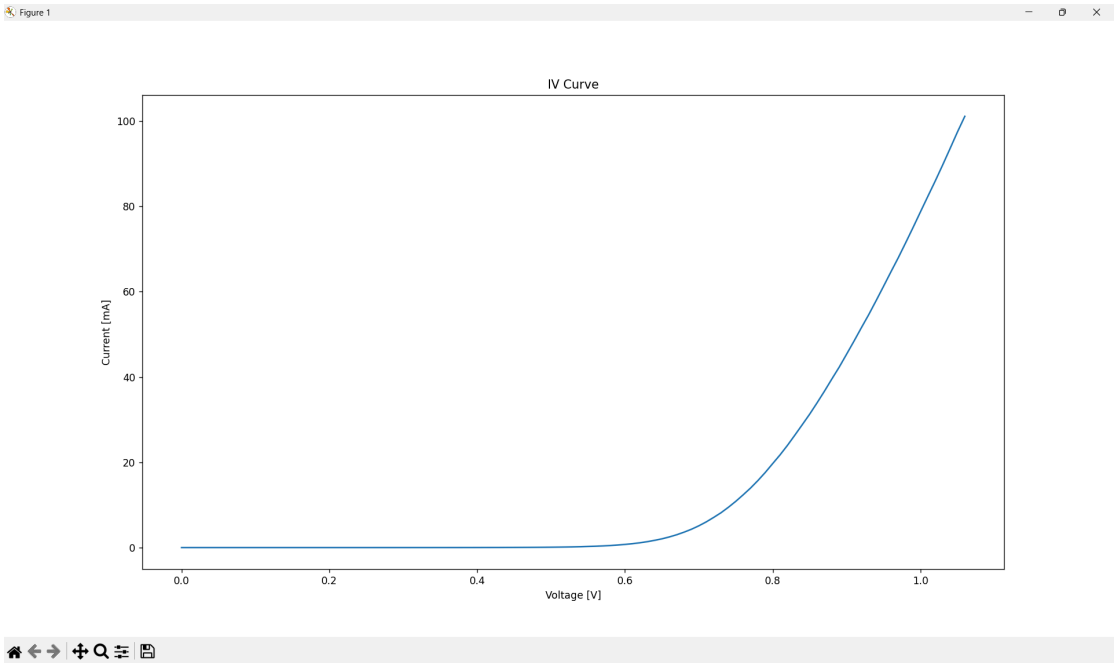


Figure 2. I-V Curve (Linear)

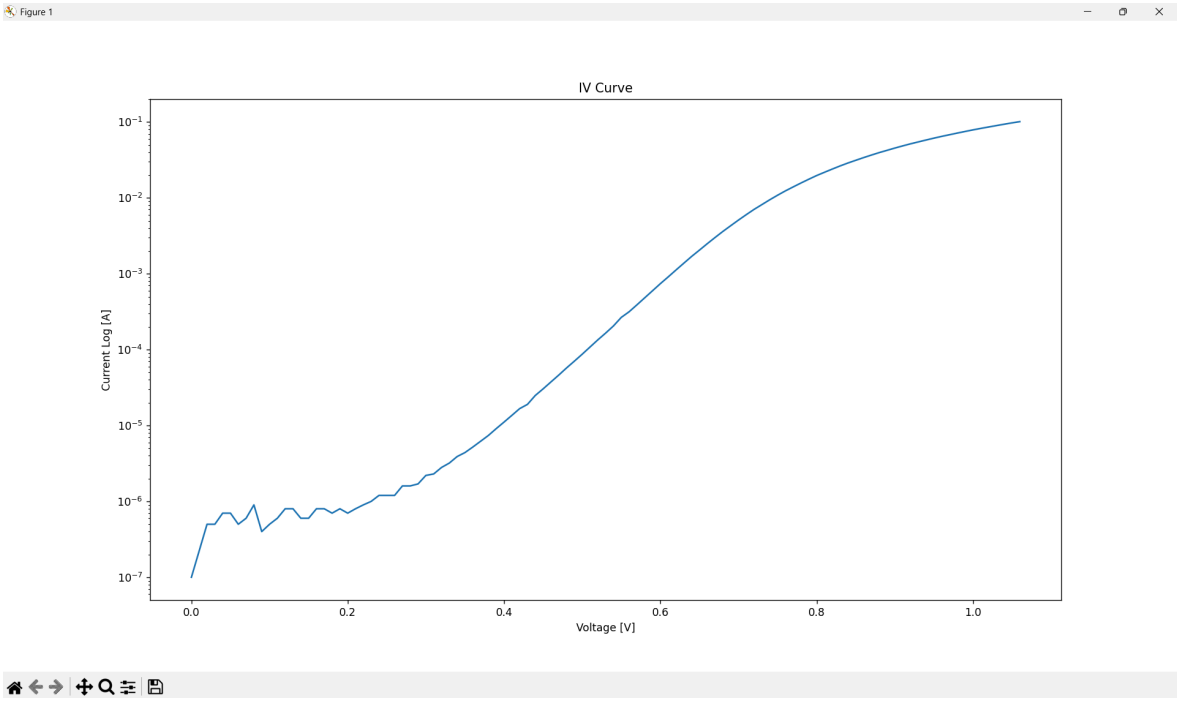


Figure 3. I-V Curve (Logarithmic)

5. Curve Fitting

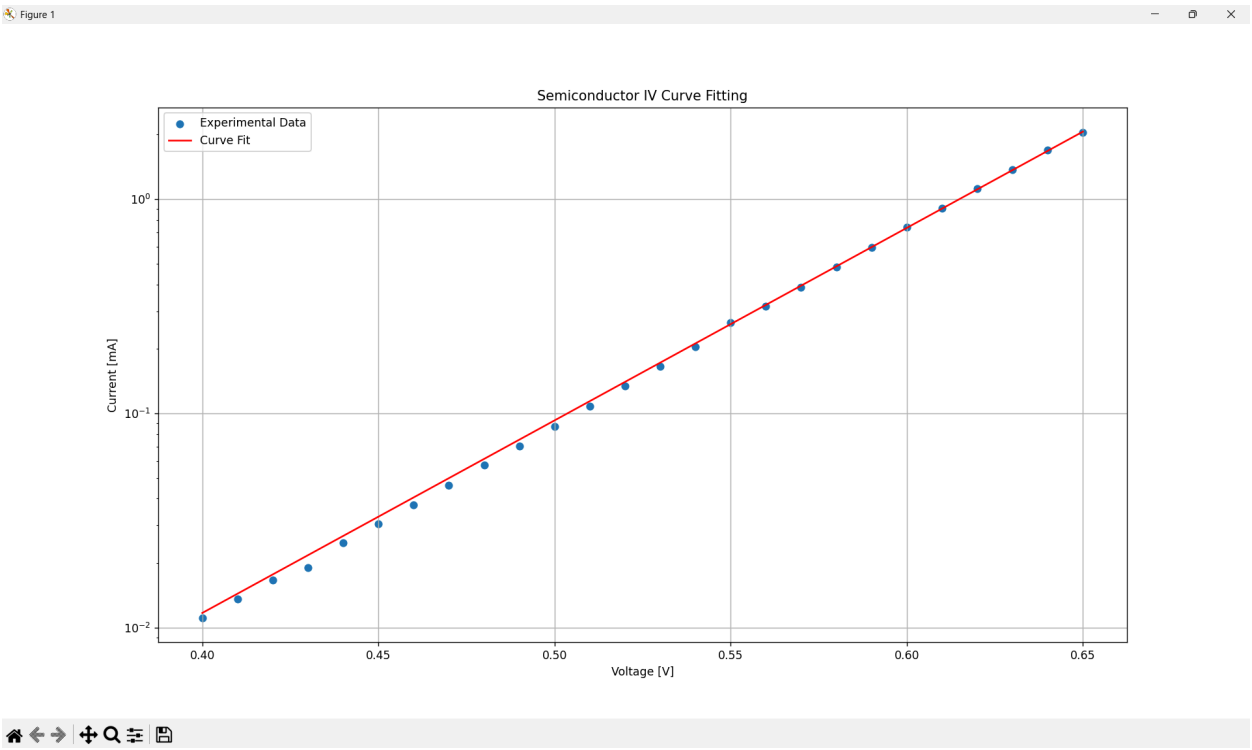


Figure 4. Curve Fitting for n and Is

```
Is = 2.9730303629345866e-09 A +- 1.4929862575446 64e-10
n = 1.8674256942817924 +- 0.007181626155702793
Logarithmic R^2 = 0.9988799683629217
```

Figure 5. Calculated Values for n and Is

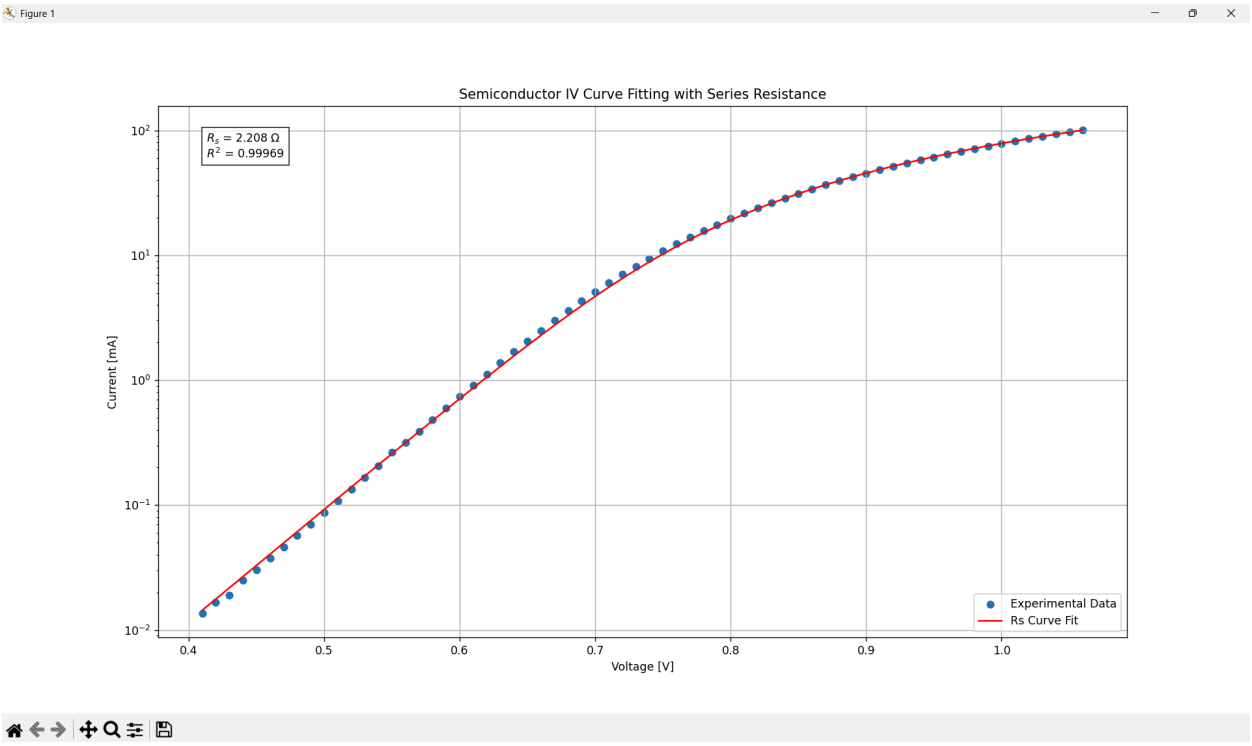


Figure 6. Curve Fitting for Rs

```
Rs = 2.207621937224117 Ohms +- 0.0025702104647974 76
Logarithmic R^2 for Rs fit = 0.9996870690327874
```

Figure 7. Calculated Values for Rs

6. Extracted Diode Graphs

In the last section, these values were derived:

- Is = 2.97e-9 A
- n = 1.867
- Rs = 2.21 Ohms

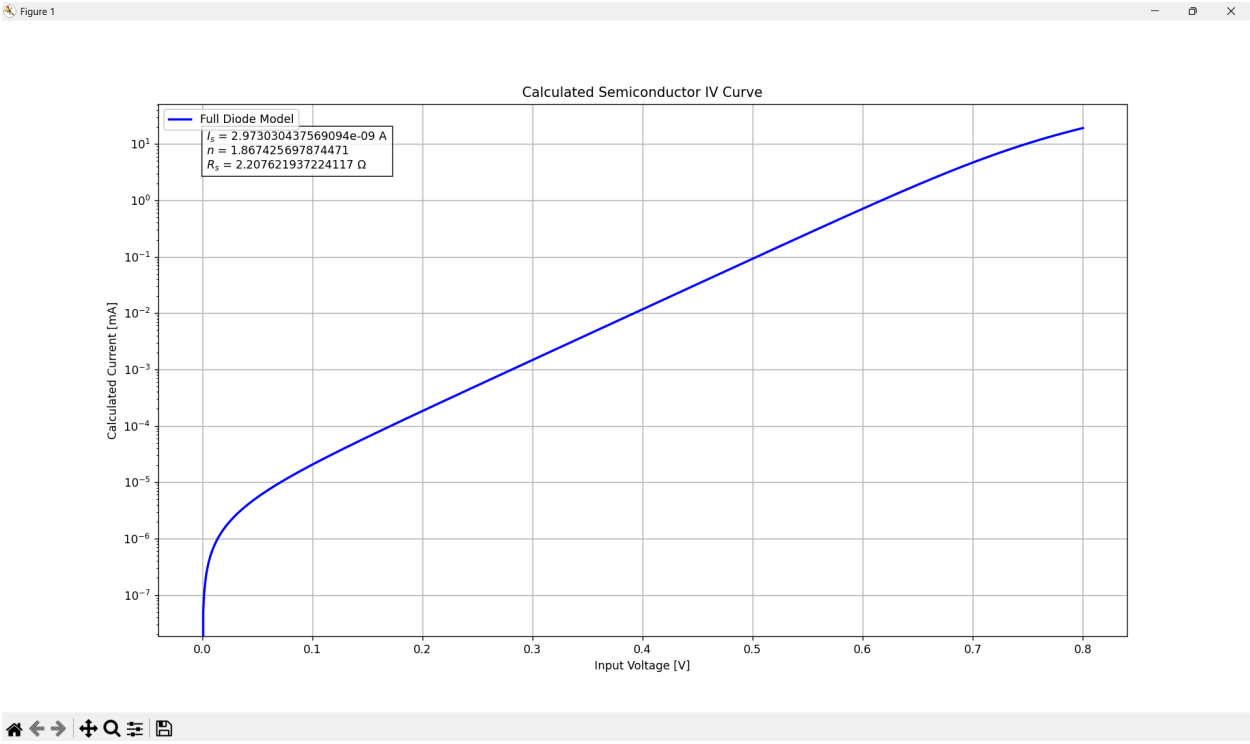


Figure 8. Calculated Standalone Diode Graph

Voltage	Current (mA)
0	
0.1	
0.2	
0.3	
0.4	
0.5	
0.6	

0.7	
0.8	
0.9	
1	

Table 2. Table of Calculated Values

We can compare this with the experimental data to provide an understanding on whether our model is accurate:

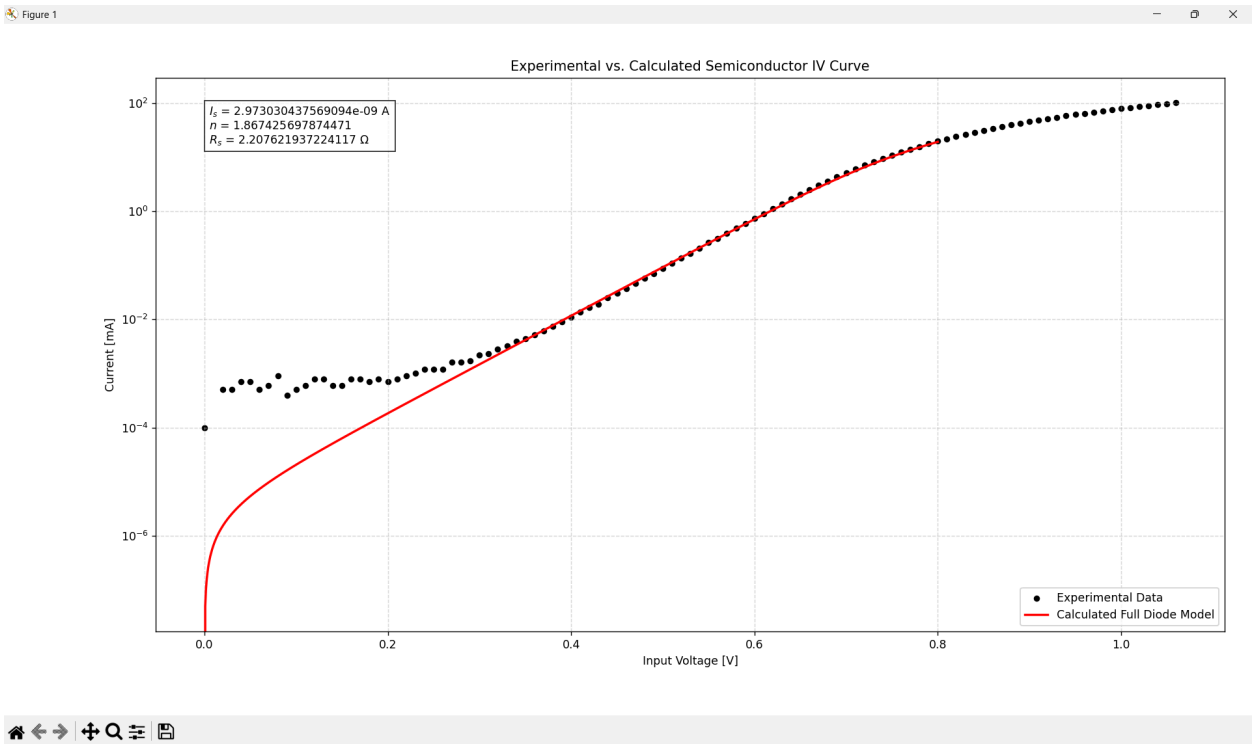


Figure 9. Experimental vs Calculated IV Curve

What we notice is that for the middle region, we have a very accurate estimate of the data that we see. However, because of the limitations of the equipment, we find that the data diverges significantly from one another at the lower end. Furthermore, the diode model only accounts up to 0.8V, while the experimental goes past. Narrowing the focus does again show the accuracy being quite high for our calculated values of Is, n, and Rs.

7. Diode Spice Simulation

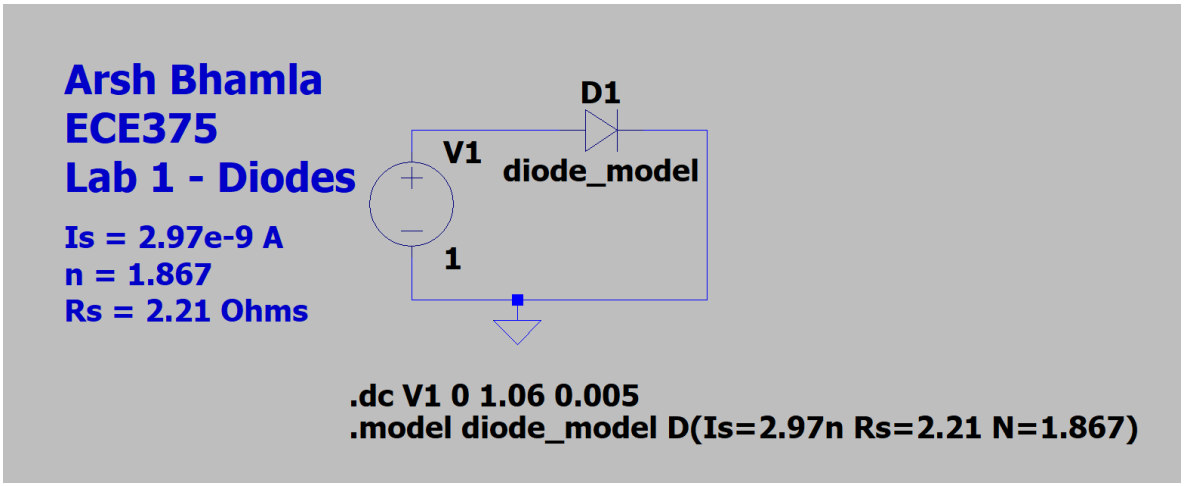


Figure 10. LTSpice Simulation with Custom Values

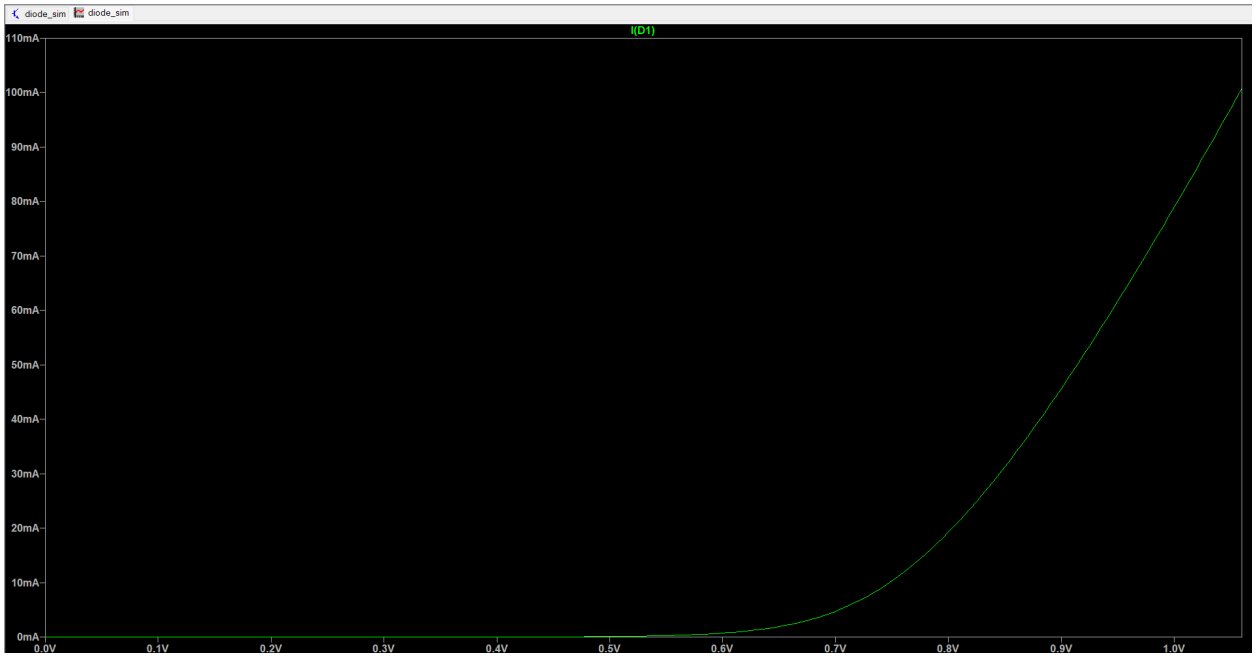


Figure 11. LTSpice Simulation Results

We can see a sample of the values:

Voltage	Current (mA)
0	0

0.1	2.06E-05
0.2	1.84E-04
0.3	1.48E-03
0.4	1.17E-02
0.5	9.28E-02
0.6	7.16E-01
0.7	4.72E+00
0.8	1.93E+01
0.9	4.57E+01
1	7.89E+01

Click here for the table

Adding the third plot in, we get all three plots laid over one another to see how accurate our values are:

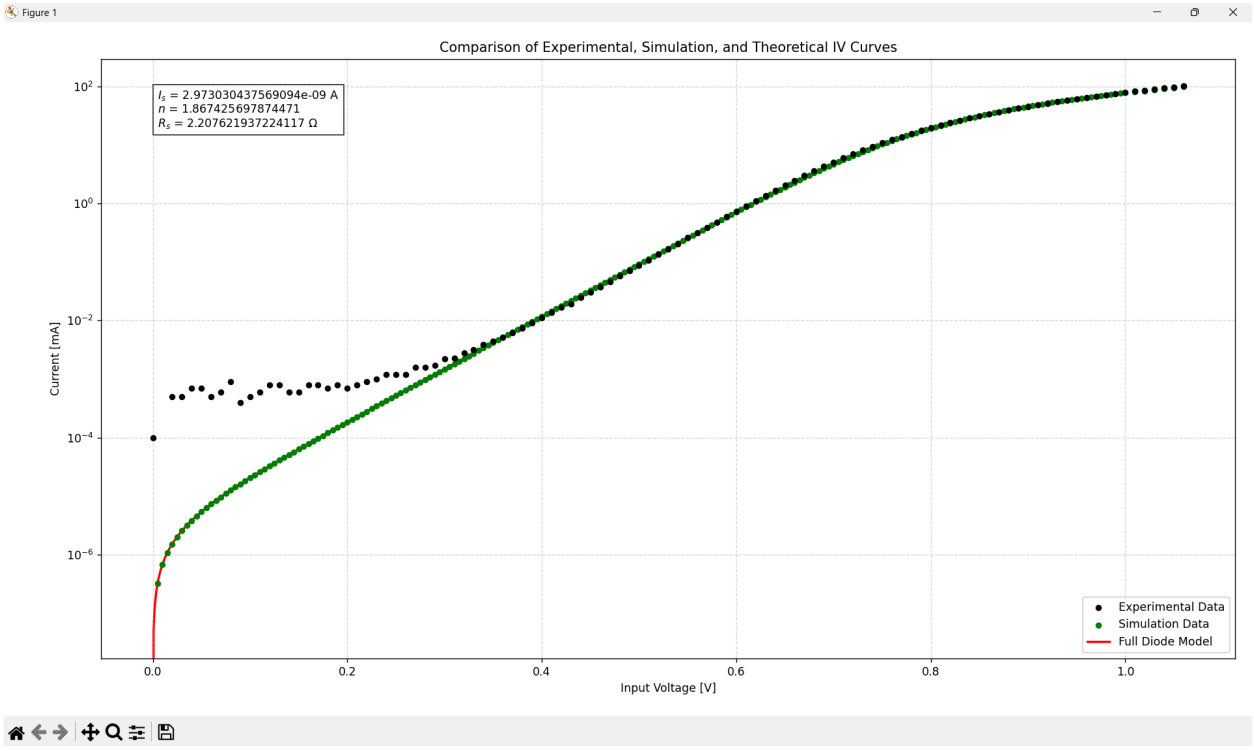


Figure 12. Measured, Calculated, and Spice Simulated IV Curves

The black dots are the original measured values. The red is the diode model I calculated using the values in Python. And the green is the SPICE simulation using those same Is, n, and Rs values. This provides the proof of accuracy between the models, while the measured values do diverge on the lower end. The data was not trained to include that which maintains the fidelity of the rest of the curve.

8. Current Fluctuations

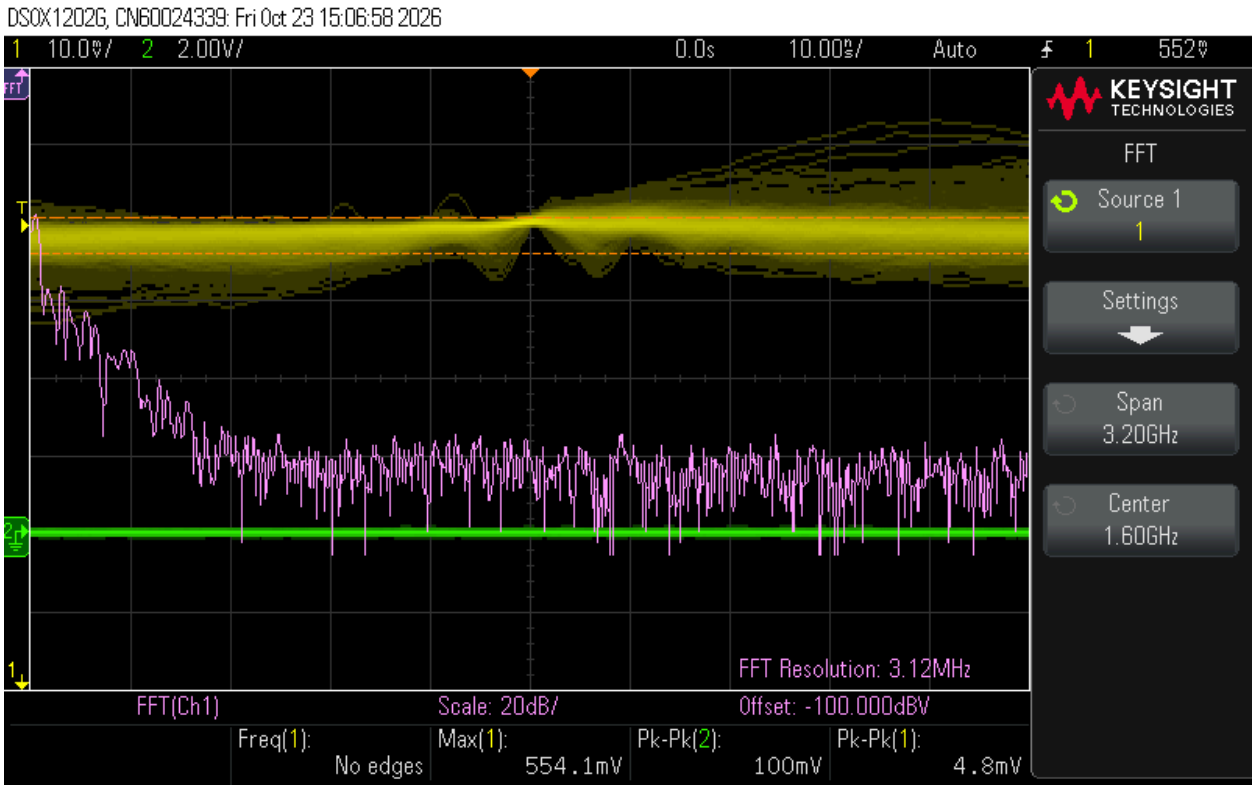


Figure 13. Oscilloscope Reading

We have both a time domain function in yellow and the fast fourier transform (FFT) in purple. The FFT is the important part since we were capturing noise; we notice that the noise power is dominated by the lower frequencies, and evens out at higher frequencies. This is characteristic of flicker noise, which has 1/f frequency dependence. At higher frequencies, we are most likely seeing the thermal noise of the system, which has uniform power at all frequencies.

9. Explanations and Conclusion

This section will cover the explanation of the results, as well as the conclusion portion for the lab report.

We looked at three main plots in Section 7, where we saw the differences between the experimental data, the calculated data, and lastly, the simulated data from LTSpice. The models (calculated and simulated) look pretty accurate above 0.4. The reason it has this divergence comes from how we found the values.

Looking at the python:

```python
def main():

    # curve fitting changed to numpy
    V_data = df["Voltage"].to_numpy()
    I_data = df["Current mA"].to_numpy() / 1000

    # use only the linear region
    mask = (V_data >= 0.4) & (V_data <= 0.65)
    V_data_lin = V_data[mask]
    I_data_lin = I_data[mask]

    # set up my initial guesses
    initial_guess = [1e-9, 1.5]

    # performing curve fit
    popt, pcov = curve_fit(
        diode_model1,
        V_data_lin,
        I_data_lin,
        p0=initial_guess,
        bounds=([0, 0.5], [np.inf, 3]),
        maxfev=20000
    )

    Is_fit, n_fit = popt
    uncertainties = np.sqrt(np.diag(pcov))

    print(f"Is = {Is_fit} A +- {uncertainties[0]}")
    print(f"n = {n_fit} +- {uncertainties[1]}")
```

Figure 14. Is and n Curve Fitting

A brief explanation of the code should start. We start off by extracting the data from a csv file where it was saved. Then we apply a mask, which only uses a portion of the full experimental data. For this, I chose to split my curve fitting to stay within the linear region to not have the effects of Rs come into play.

To give context for the rest of the code shown, I next have to set up my initial guesses to give it a start, and since both values had an expected range I was able to start them closer to their actual result.

The curve fitting model on python uses a library called scipy.optimize, and the next function uses the basic diode model and the data to find the results.

Lastly, we plot this as a point and uncertainty, and below this (not shown) is the R^2 value for confidence scoring.

```
# V_app = I*R_s + n*Vth*ln(I / I_s + 1)
initial_guess_Rs = [1]

# disregards the bottom bad part of the data
mask = V_data > 0.4
V_data_r = V_data[mask]
I_data_r = I_data[mask]

popt_Rs, pcov_Rs = curve_fit(
    diode_model2,
    V_data_r,
    I_data_r,
    p0=initial_guess_Rs,
    bounds=([0.001], [1000]),
    maxfev=20000
)

Rs_fit = popt_Rs[0]
Rs_uncertainty = np.sqrt(np.diag(pcov_Rs))[0]

print(f"Rs = {Rs_fit} Ohms +- {Rs_uncertainty}")
```

Figure 15. Rs Curve Fitting

This is the calculation of Rs, and I hard-coded the values of Is and n into this portion since it was found in a previous step.

You can find that we again used the masking, this time to only past 0.4V, which explains the reason why the data has less accuracy in that region.

The full code used can be found here

To summarize, we were able to deduce and understand

10. Additional IV Curves

Temperature has a major effect on the diode parameters.

The first and most obvious parameter is the thermal voltage, $V_{th} = \frac{k_B T}{q}$

Looking at the formula for saturation current,

$I_S = Aqn_i^2 \frac{D_n}{L_n N_A}$

Breaking down for each parameter:

A is area, which is not temperature dependent

q is charge of an electron, also not temperature dependent

$n_i^2 = \sqrt{N_C N_V} e^{-\frac{E_G}{2kT}}$, where N_C, $N_V \propto T^{\frac{3}{2}}$, providing $n_i^2 \propto T^{\frac{3}{2}}$

n_i represents the intrinsic carrier concentration of silicon. Temperature increases the amount of electron-hole pairs being generated

$D_n = \frac{kT}{q}\mu_n$, where μ_n has a temperature dependence dependent on the carrier concentration. The datasheet does not include information about the carrier concentration, so I will assume a moderately doped Si with $10^{16} cm^{-3}$ (from the 1400 value at room temperature that we use in class). This provides a negative temperature dependence for mobility.

$\mu_{n,\,250K} \approx 2000$

$\mu_{n,\,375K} \approx 1000$

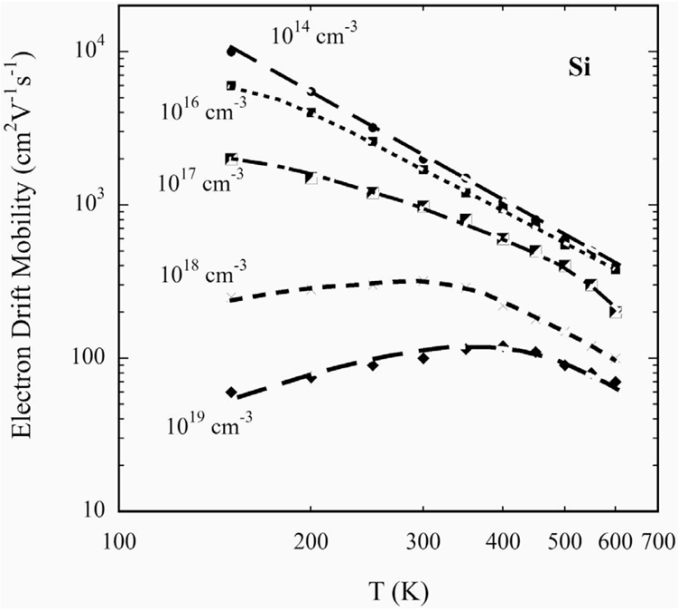


Figure 16. Mobility vs Temperature

D_n is the diffusion coefficient for electrons. The two terms combined give a weak temperature dependence since one is a positive temperature dependence and the other is negative.
L_n is the average length before collision, which has some temperature dependence. As temperature increases, collisions increase as well, but it should be negligible relative to the other components.
N_A is the acceptor concentration, which is not temperature dependent.

What we determine is that the n_i^2 component provides a $T^{\frac{3}{2}}$ dependence, and every other component in comparison has negligible changes.

Implementing this into code, the diode model was changed to include temperature dependence. the parameters derived are:

```
Temperature-dependent diode parameters
T1 = 375 K
Is(T1) = 6.5330726975814e-05 A
Vth(T1) = 0.032343750000000004 V

T2 = 250 K
Is(T2) = 1.6338771144562 98e-13 A
Vth(T2) = 0.021562500000000002 V

n = 1.867425697874471
Rs = 2.207621937224117 Ohms
```

Figure 17. Diode Parameters Dependent on Temperature

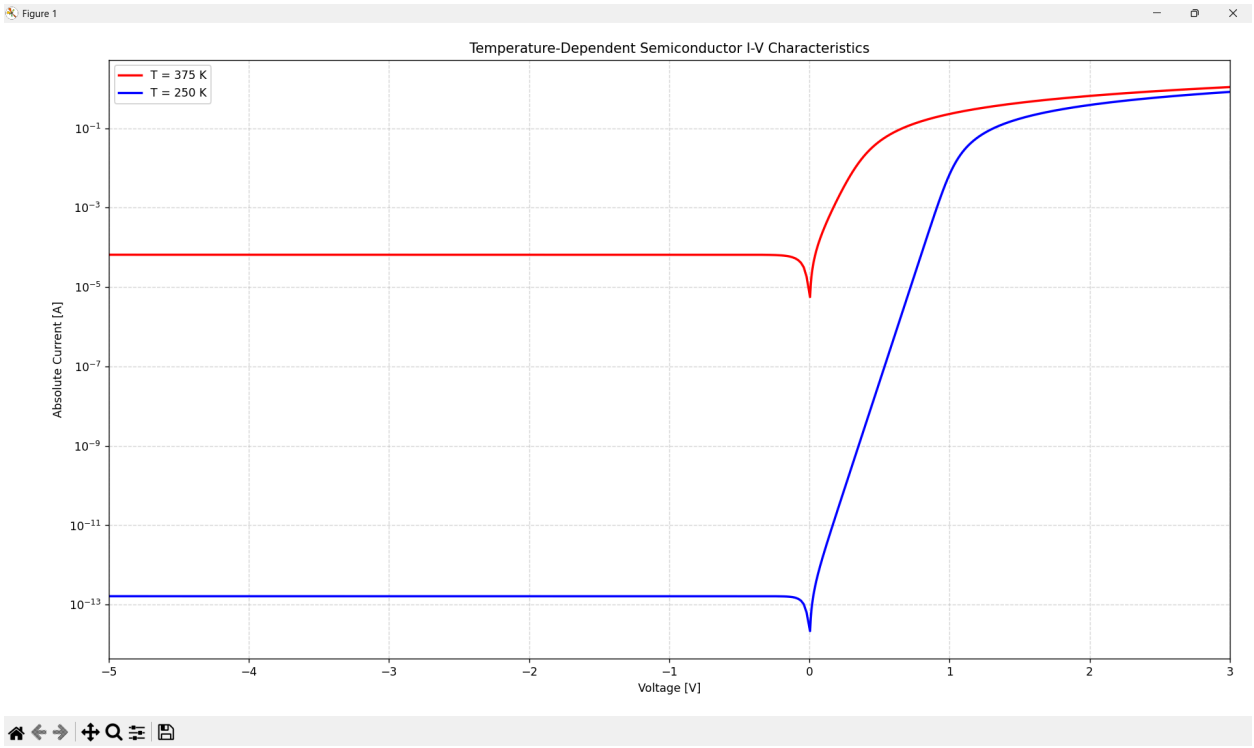


Figure 18. I-V Curve Based on Temperature

Questions

1. Can your experimental data be fitted using the empirical diode equation?
The diode equation fits the experimental data for the majority of the plot. However in earlier portions of the IV curve, the equipment saturates in its ability to read lower current measurements, and so we see a divergence.

2. Are there any significant discrepancies between your experimental data, calculations, and Multisim simulations (i.e., I-V characteristics, the values of saturation current Isat, diode ideal factor n, and series resistor Rser)?
The earlier portion of the experimental data has a discrepancy, but the calculation and Multisim simulations are line on line

3. If yes, how can these discrepancies be explained?
The limitations of the test equipment and test setup in measuring smaller currents getting closer to 0V (which should be equal to the Is) means that there is a discrepancy.

4. What is most likely the origin of the diode noise?
We see higher power in lower frequencies, which is the power spectral density of flicker noise (also known as 1/f noise).